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Lab 1

Windows Command Prompt

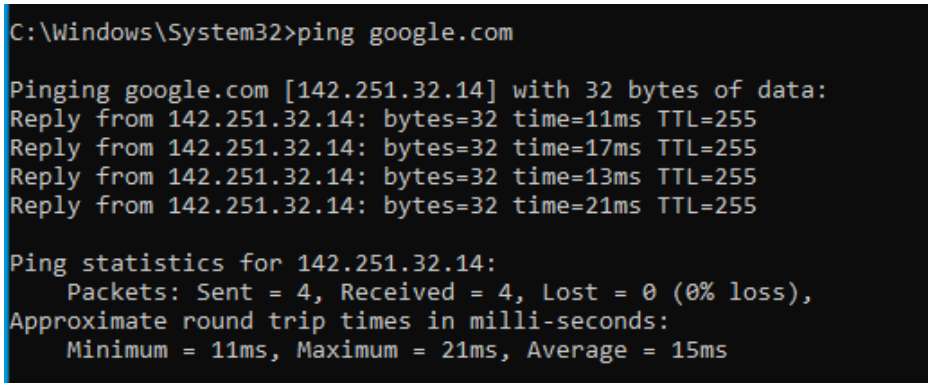
Non-technical Summary

In this lab we explored Windows command prompt and some of its commands. Command prompt is a tool that is used for communicating with the operating system. Command prompt is a built in application in the Windows OS. It uses a command line interface (CLI) instead of a graphical user interface (GUI). Commands are very straightforward, and they give the user the ability to do things such as file and directory management, ping a network, access services such as task lists, or scan and repair file system errors. Even though the GUI is more user friendly, it is common for command prompt to help users with troubleshooting issues that do not show up in the GUI. Command prompt also makes automating processes through scripts possible. Once familiar with the commands, command prompt is a simple and easy way for a user to achieve a quick fix, and manage the computer's system in general.

Technical Report

Command prompt can be started by searching "cmd" or "command prompt" in the start menu. The beginning of the lab covers some basic commands, such as changing working directories, file management, and displaying date and time. As you can see below, many of these commands are quite simple and easy to remember. Like many other commands in command prompt, these commands can also be built upon to create more complex commands.

Some important network commands include ping and tracert. Ping tests the connection of a network between a host and IP, for example, your computer and google.com. As we can see ping outputs the host IP, number of bytes sent, reply time in milliseconds, and time to live (TTL).



We can modify the ping command by using -t to specify the number of pings to send, -i to set an interval between each ping, or -4 and -6 to specify IPv4 or IPv6 respectively.

Tracert is similar to ping in that it also runs a connection test from one device to another. However, tracert finds the exact route taken, and time taken at each “hop” to reach the server. The first column is the hop number, the next three columns are the time in milliseconds it took to attempt to send the data at each hop, or round trip time (RTT).

```
C:\Windows\System32>tracert google.com

Tracing route to google.com [142.250.190.142]
over a maximum of 30 hops:

  1    14 ms    12 ms    15 ms  ord37s36-in-f14.1e100.net [142.250.190.142]

Trace complete.
```

We can modify the tracert command by using -d to avoid resolving hostnames, -h to change maxhops, -w set timeout.

Some important services commands include tasklist and taskkill. Tasklist displays all current running processes on a device. Tasklist displays image name, PID, session name, session#, and memory usage.

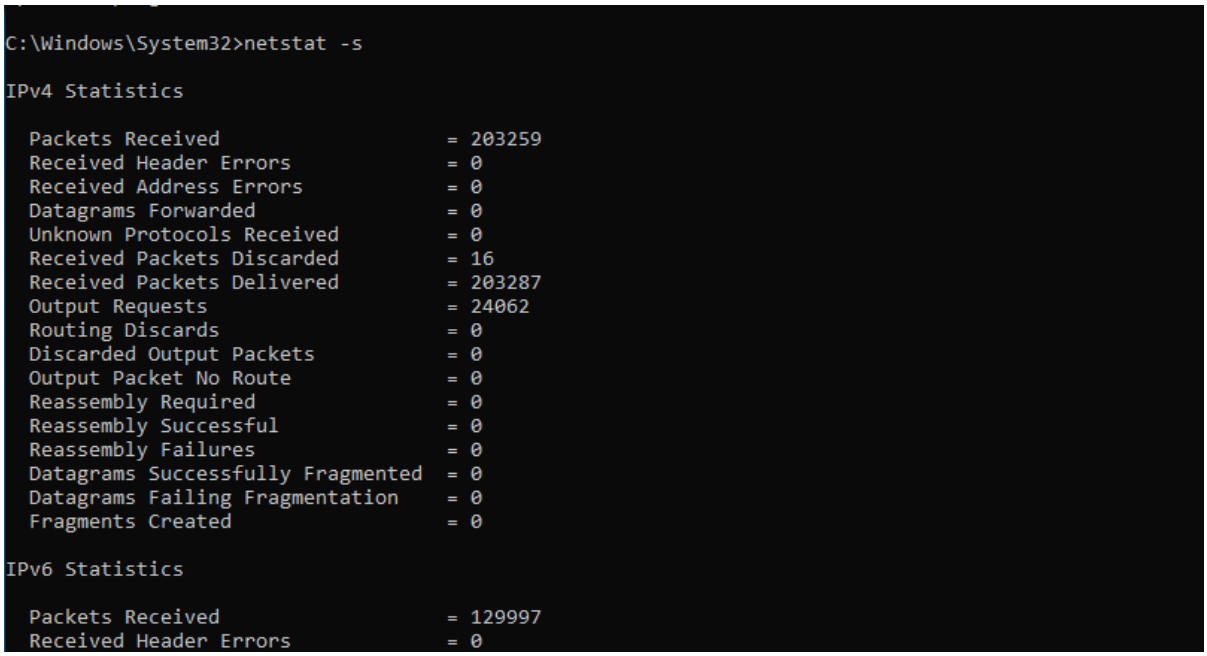
```
C:\Windows\System32>tasklist

Image Name                     PID Session Name        Session#    Mem Usage
========================= ======== ================ =========== ============
System Idle Process              0 Services                   0          8 K
System                           4 Services                   0        136 K
Registry                        72 Services                   0     15,232 K
smss.exe                       308 Services                   0        956 K
csrss.exe                      404 Services                   0      4,596 K
wininit.exe                    472 Services                   0      6,156 K
csrss.exe                      480 Console                    1      5,916 K
winlogon.exe                   540 Console                    1      9,300 K
services.exe                   564 Services                   0      7,396 K
lsass.exe                      572 Services                   0     18,840 K
svchost.exe                    684 Services                   0     21,948 K
fontdrvhost.exe                692 Services                   0      2,440 K
fontdrvhost.exe                752 Console                    1      3,428 K
svchost.exe                    796 Services                   0     14,176 K
dwm.exe                        884 Console                    1     45,048 K
svchost.exe                    984 Services                   0     61,492 K
```

We can modify the tasklist command by using /svc to display services hosted in each process listed.

Taskkill is used to terminate processes listed by PID as in the tasklist, or .exe file. This can be very useful in situations where troubleshooting cannot be easily and efficiently done via the GUI. Taskkill is especially useful in situations where programs or processes are not responding, and a force quit is required.

Netstat is a command that shows us network status, and details of current TCP/IP connections. We can modify the command using -a to display connections and listening ports, -e for ethernet stats, and -s for additional stats for TCP, UDP, TCPv6, and others.



Chkdsc is an easy way to check the status of a disk, it scans it for errors in systems or in sectors. Chkdsc can also repair these errors which aids in protecting data from being lost, using /f and /r. We are able to see the size of each system and allocation.

Driverquery displays the installed drivers on a system. The output of driverquery is module name, display name, driver type, and link date. Driverquery gives us more information about what each driver does as well as how they communicate with the operating system.

```
C:\Windows\System32>driverquery

Module Name  Display Name           Driver Type   Link Date
============ ====================== ============= ======================
1394ohci     1394 OHCI Compliant Ho Kernel
3ware        3ware                  Kernel        5/18/2015 5:28:03 PM
ACPI         Microsoft ACPI Driver  Kernel
AcpiDev      ACPI Devices driver    Kernel
acpiex       Microsoft ACPIEx Drive Kernel
acpipagr     ACPI Processor Aggrega Kernel
AcpiPmi      ACPI Power Meter Drive Kernel
acpitime     ACPI Wake Alarm Driver Kernel
Acx01000     Acx01000               Kernel
ADP80XX      ADP80XX                Kernel        4/9/2015 3:49:48 PM
```

Queryex can be thought of as an advanced version of sc query. We can query specific services, such as eventlog. Here we get details of the service name, type, PID, error code, time needed to complete an operation, and checkpoint during operation such as start, stop, or pause.

```
SERVICE_NAME: eventlog
        TYPE               : 20  WIN32_SHARE_PROCESS
        STATE              : 4  RUNNING
                                (STOPPABLE, NOT_PAUSABLE, ACCEPTS_SHUTDOWN)
        WIN32_EXIT_CODE    : 0  (0x0)
        SERVICE_EXIT_CODE  : 0  (0x0)
        CHECKPOINT         : 0x0
        WAIT_HINT          : 0x0
        PID                : 1004
        FLAGS              :
```

Another great command is whois, using Sysinternals. The whois command requires a domain name or IP, then gives details such as the names of servers, domain records and dates. It will also give information about who is registered to the domain, such as name address, phone number, etc.

Also using Sysinternals, we can use psgetsid. This command gets us the security identifier. This security identifier can get us to account names, and help retrieve additional information about a system such as display names. One step further, and we can use psinfo, which allows us to view detailed information such as time system has been in use, kernel, build, version, processor, and memory.